

Compound Pendulum

Aim:-

1. To Plot a graph of period of oscillations (T) vs distance (l) of point of suspension S from C.G.
2. Using the measurement from the graph, calculate g and the radius of gyration k
3. Calculate the the moment of inertia about CM. Compare the result obtained by the direct calculation $I_{CM} = \frac{ML^2}{12}$
4. Measure the period of oscillation about an arbitrary pivot point which is at a distance d from the CM. Calculate $I_{Pivot} = Mgd \frac{T^2}{4\pi^2}$
5. Verify the result using parallel axis theorem : $I_{Pivot} = I_{CM} + M d^2$

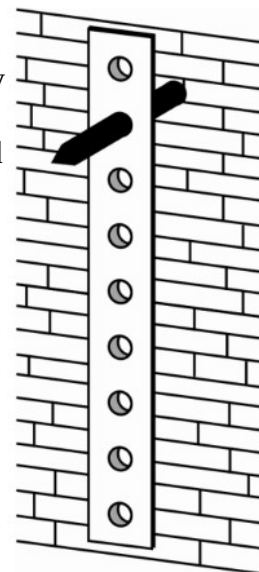
Apparatus:

- Compound pendulum
- A stop watch

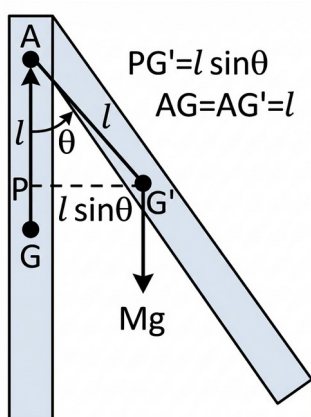
Description:

A compound pendulum is a rigid body capable of oscillating freely about a horizontal axis. In the physics laboratory, it is normally available in the form of a bar of length nearly one metre and width about 2.5 cm. A series of circular holes, 5-6 mm in radius, are drilled symmetrically about its centre of gravity (C.G.), i.e. along the length of the bar.

The centres of any two consecutive holes are at equal distances of about 5 cm. These holes allow the bar to be suspended from a knife-edge. Usually, two movable knife-edges are provided with the bar pendulum. These can be fitted successively in various holes, one on each side of C.G. and at equal distances from it.



Principle:



A= Point of Suspension
G= Centre of Gravity

Torque acting on the pendulum due to its weight Mg about the point of suspension A is

$$\tau = mgl \sin \theta$$

I = Moment of inertial of the pendulum about A

$$\therefore I \frac{d^2 \theta}{dt^2} = -mgl \sin \theta \quad \text{or} \quad \frac{d^2 \theta}{dt^2} + \frac{mgl}{I} \theta = 0 \quad \text{when } \theta \text{ is small}$$

This shows the motion is simple harmonic with period $T = 2\pi \sqrt{\frac{I}{Mgl}}$

But $I = I_{CM} + Ml^2$ and $I_{CM} = Mk^2$, k = radius of gyration about A

$$\therefore T = 2\pi \sqrt{\frac{Mk^2 + Ml^2}{Mgl}} = 2\pi \sqrt{\frac{k^2 + l^2}{gl}}$$

$$\text{or } T = 2\pi \sqrt{\frac{L}{g}} \quad \text{where } L = l + \frac{k^2}{l} \text{ is called 'Length of the equivalent Simple Pendulum'}$$

- If we take a point O on the line AG produced such that $AO = L$, then this point O is called centre of oscillations.
- If the pendulum is inverted and made to oscillate about O, then it can be proved that its time period $T' = T$

$$T = 2\pi \sqrt{\frac{L}{g}}, \quad \text{where } L = l + l' \quad \text{with } l' = \frac{k^2}{l}$$

$$\therefore T = 2\pi \sqrt{\frac{l + l'}{g}} = 2\pi \sqrt{\frac{l + \frac{k^2}{l}}{g}} \quad \text{and} \quad T' = 2\pi \sqrt{\frac{l' + \frac{k^2}{l'}}{g}}, \quad T = T'$$



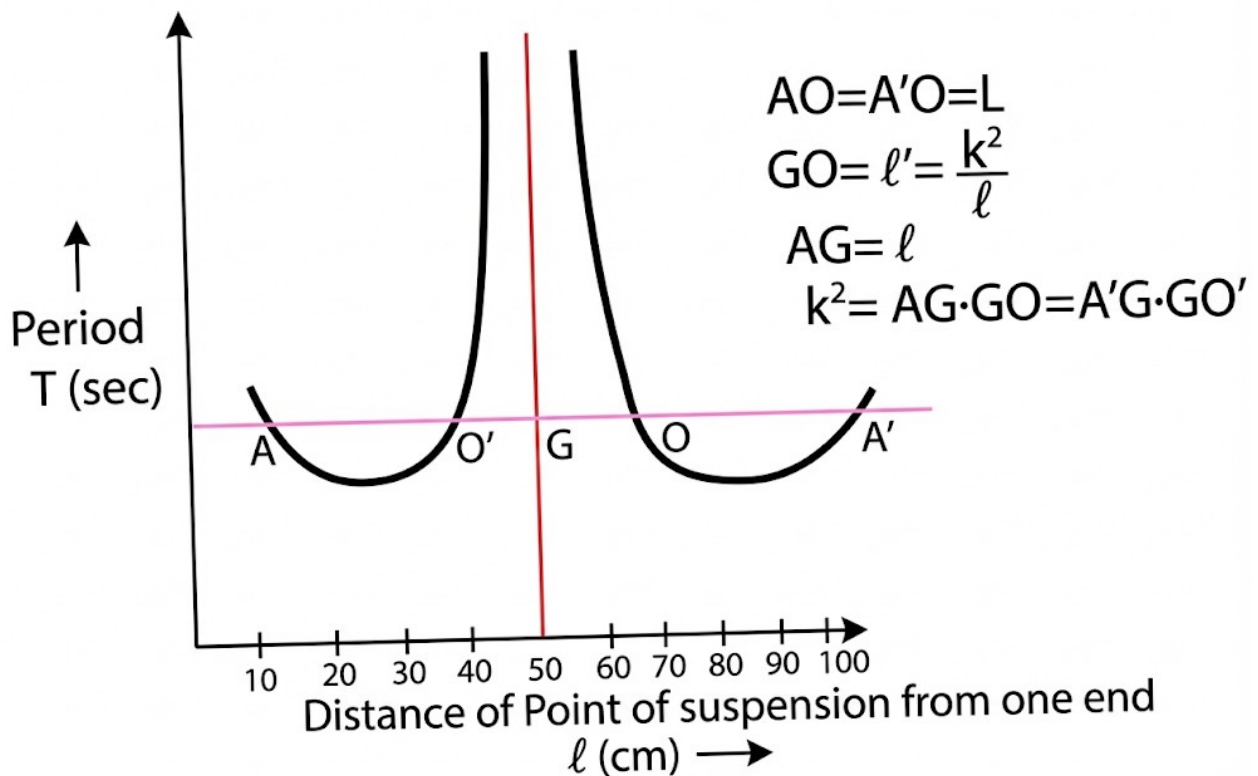
$$AG = A'O = l$$

$$GO = l' = \frac{k^2}{l} \quad GO' = l = \frac{k^2}{l'}$$

$$AO = A'O' = L = l + l'$$

$$k^2 = l l'$$

- **Symmetry by Inversion:** If we invert the pendulum and choose a suspension point **A'** that is symmetrical to **A** (equidistant from the C.G.), the period of oscillation remains constant.
- **Conjugate Points:** Just as **A** has a center of oscillation **O**, the inverted point **A'** has a corresponding center of oscillation **O'**.
- **The Four-Point Rule:** Therefore, the period **T** is the same for oscillations about four specific points:
 1. Point of suspension (A)
 2. Center of oscillation (O)
 3. Symmetrical point of suspension (A')
 4. Symmetrical center of oscillation (O')
- **Graphical Evidence:** This is observed in the **T** vs. **l** plot, where the curve is symmetrical about the **y**-axis.



Experimental Procedure

1. Data Collection:

- Begin by suspending the pendulum from the first hole (located 5 cm from the upper end).
- Measure the time taken for 20 complete oscillations t .
- Record the distance (l) from the upper end to the axis of oscillation.
- Calculate the time period of oscillation, $T = t/20$.

2. Iteration:

- Repeat the measurement for all subsequent holes along the bar.
- Note: Skip the hole located at the Center of Gravity (C.G.), as oscillations are not possible at this point.
- Tabulate all values of distance (l) against the corresponding time period (T).

3. Graphical Analysis:

- Plot a graph of l (x-axis) versus T (y-axis).
- From the graph, determine the acceleration due to gravity (g) and the radius of gyration (k).

4. Moment of Inertia Calculation:

- Measure the total mass (M) of the pendulum.
- Calculate the Moment of Inertia about the Center of Mass (I_{CM}) using the derived value of k (where $I_{CM} = Mk^2$).

5. Verification:

- Calculate the Moment of Inertia (I_{Pivot}) about a specific axis located at a distance d from the C.G.
- Verify this result using the Parallel Axis Theorem: $I_{Pivot} = I_{CM} + M d^2$

Observations:

Mass of the Pendulum = M= . . .

1. To draw l vs T graph:

Trial No.	Distance of the point of suspension from the upper end l (cm)	Time t for 20 oscillations in seconds			Period T (Seconds)
		1	2	Mean	
1	5				
2	10				
3	15				
4	20				
5	25				
6	30				
7	35				
8	40				
9	45				
10	55				
11	60				
12	65				
13	70				
14	75				
15	80				
16	85				
17	90				
18	95				
19	100				

2. To find g and L from graph:

Trial No.	Length of the equivalent simple pendulum L (m)		Mean L (m)	Period T (seconds)	$g = 4\pi^2 \frac{L}{T^2}$
	AO	A'O'			

Mean acceleration due to gravity $g = \dots\dots\dots$

3. To find the radius of gyration k from graph:

Trial No.							Mean k (m)
	AG(m)	GO (m)	k=AG.GO	A'G(m)	GO'(m)	k=A'G.GO'	

The distance between the minima of the curve = $2k = \dots$

$\therefore$ Mean Value of Radius of gyration $k = \dots$

4. To find the moment of inertia I_{CM} (from Radius of Gyration):

Mass of the Pendulum = $M = \dots$

Radius of gyration = $k = \dots$

$$I_{CM} = Mk^2 = \dots$$

5. To find the moment of inertia I_{CM} (By Direct Method):

Mass of the Pendulum = $M = \dots$

Length of the Pendulum = $L_0 = \dots$

$$I_{CM} = ML_0^2/12 = \dots$$

6. To find the Moment of Inertia about an arbitrary pivot point I_{pivot} and verify Parallel Axis Theorem :

Moment of inertia about the centre of mass $I_{CM} = \dots$

Trial No.	Distance of the Pivot Point from the C.G. d (m)	Period in Seconds			I_{Pivot}	
		On Positive side of C.G.	On Negative side of C.G.	Mean Period T(second s)	By Direct Calculation $I_{Pivot} = Mgd \frac{T^2}{4\pi^2}$	By Parallel Axis Theorem $I_{Pivot} = I_{CM} + Md^2$
1	0.2					
2	0.3					
3	0.4					

Results:

1. Acceleration due to gravity at the place $\mathbf{g} = \dots\dots$
2. Radius of gyration of the pendulum about an axis through C.G. $= \dots\dots$
3. Moment of inertia of the pendulum about an axis passing through C.G and perpendicular to its length $\mathbf{I_{CM}}$:
From Radius of Gyration , $\mathbf{I_{CM}} = \dots\dots$
From Direct Method , $\mathbf{I_{CM}} = \dots\dots$
4. Moment of Inertia about an arbitrary pivot point $\mathbf{I_{pivot}}$ is calculated by direct method and verified using Parallel axis theorem
5. The Moment of Inertia about an arbitrary pivot point $\mathbf{I_{pivot}}$ is initially determined using the direct calculation. This value is subsequently verified for accuracy by applying the Parallel Axis Theorem.